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Ecosystem Orchestration: Mechanisms and Regimes

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Abstract:	Value creation is increasingly shifting from standalone firms in linear value chains toward multilaterally coordinated efforts among heterogeneous actors in ecosystems. Yet, we know relatively little about how coordination unfolds when value creation activities are distributed in this manner. As a focal entity grows increasingly reliant on actors beyond its formal authority, value realization depends on aligning and steering these interdependent actors through indirect means—an approach best described as orchestration. We define ecosystem orchestration as the strategic application of control and coordination mechanisms beyond organizational boundaries to integrate complementary efforts across ecosystem participants. First, through a comprehensive integrative review, we identify four categories of mechanisms: codified, cognitive, incentive, and hierarchical. We then advance a typology by examining how these mechanisms are bundled into distinct orchestration regimes across platform, industrial, innovation, and entrepreneurial ecosystems. Further, we highlight salient connections between these mechanisms and theories of the firm—RBV, PRT, TCE, and KBV. By moving beyond generic notions of orchestration and revealing its underlying mechanisms, we establish a clearer foundation for advancing ecosystem research, integrate the fragmented ecosystem orchestration literature, and anchor it more firmly in established theories. Finally, we conclude by outlining conceptual, empirical, and methodological directions for future inquiry.

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ABSTRACT

Value creation is increasingly shifting from standalone firms in linear value chains toward multilaterally coordinated efforts among heterogeneous actors in ecosystems. Yet, we know relatively little about how coordination unfolds when value creation activities are distributed in this manner. As a focal entity grows increasingly reliant on actors beyond its formal authority, value realization depends on aligning and steering these interdependent actors through indirect means—an approach best described as orchestration. We define ecosystem orchestration as the strategic application of control and coordination mechanisms beyond organizational boundaries to integrate complementary efforts across ecosystem participants. First, through a comprehensive integrative review, we identify four categories of mechanisms: codified, cognitive, incentive, and hierarchical. We then advance a typology by examining how these mechanisms are bundled into distinct orchestration regimes across platform, industrial, innovation, and entrepreneurial ecosystems. Further, we highlight salient connections between these mechanisms and theories of the firm—RBV, PRT, TCE, and KBV. By moving beyond generic notions of orchestration and revealing its underlying mechanisms, we establish a clearer foundation for advancing ecosystem research, integrate the fragmented ecosystem orchestration literature, and anchor it more firmly in established theories. Finally, we conclude by outlining conceptual, empirical, and methodological directions for future inquiry.

INTRODUCTION

A firm controls more than it owns, as its influence extends beyond legal boundaries (Santos & Eisenhardt, 2005). This notion appears particularly pertinent in today's ecosystem economy. As Iansiti and Levien (2004: 78) note: "*A company's performance is increasingly dependent on the firm influencing assets outside its direct control*." However, for a long time, organization and management research overlooked this phenomenon. Early studies predominantly focused on managerial authority in hierarchies and price signals in markets, neglecting the many hybrid modes of organizing economic activity. Yet, as we are now beginning to recognize, the "invisible hand" of the market and the "visible hand" of the hierarchy are not mutually exclusive, nor are they the only mechanisms that pervade economic organization (Altman, Nagle, & Tushman, 2022; Chen, Tong, Tang, & Han, 2022; Langlois, 2003). Rather, different sets of mechanisms are often required to coordinate ecosystems of actors which are characterized by local autonomy but are interdependent in their realization of a common value proposition (Adner, 2017; Altman et al., 2022; Shipilov & Gawer, 2020). We thus define ecosystem orchestration as *the strategic application of control and coordination mechanisms to align the interdependent activities of ecosystem actors*.

Ecosystem orchestration is a relatively recent topic when examined within the broader history of management research. Studies on interorganizational forms, where actors closely coordinate their activities toward a mutual goal, have emerged around late 1980s (Langlois & Robertson, 1992; Miles & Snow, 1986). These forms were primarily driven by emerging technologies, particularly information technologies, which enabled new ways to coordinate activities and reconfigure business models (Afuah, 2003; Altman et al., 2022; Porter, 2001). Essentially, these ecosystems represent a dynamic opposite to that observed by Chandler in the rise of the industrial enterprise of the late 19th century. During the Industrial Era, achieving minimum efficient scale required vertical integration to ensure close coordination of increasingly complex technological processes (Chandler, 2004: 24). While the coordination technologies of that time, such as the telegraph and the railroad, made internalization economically reasonable, this pattern began to shift with the advent of modern digital communication technologies, which enabled low-cost and instantaneous information exchange and monitoring between parties (Forman & McElheran, 2025). Firms could now concentrate on their core competencies without resorting to vertical integration and its accompanying hierarchical structures and costly bureaucratic processes (Langlois, 2003). As a result, the integration of effort—even in technologically complex processes—became possible *across* interdependent firms (Argyres, 1999). Today this phenomenon of mutual complementarity and interdependence eventually led to the coining of the term "business ecosystem," drawing a parallel to natural ecology, where species depend on and complement one another (Iansiti & Levien, 2004; Moore, 1993). This term has now become an integral part of management literature, prompting a surge of studies aimed at defining and exploring the ecosystem phenomenon, underscoring its growing significance.

Broadly defined, a business ecosystem is an *"alignment structure of the multilateral set of partners that needs to interact in order for a focal value proposition to materialize"* (Adner, 2017: 47). Essentially, these interorganizational forms are underscored by mutual interdependence and complementarity, resulting in synergy—a system greater than the sum of its parts (Cusumano & Gawer, 2002; Dhanaraj & Parkhe, 2006). A *lead firm*,[1] or a collective of such firms, must purposefully devise conditions that would align heterogeneous ecosystem actors (Cusumano &

[1] Note that several synonymous terms are used in the literature, including keystone firms (Iansiti & Levien, 2004), hub firms (Dhanaraj & Parkhe, 2006), orchestrators (Cozzolino & Geiger, 2024), and lead firms (Jacobides, Cennamo, & Gawer, 2018), among others.

Gawer, 2002; Iansiti & Levien, 2004; Jacobides, Cennamo, & Gawer, 2018). These orchestration activities are carried out by lead firms largely without direct formal control over the actors (Adner, 2017; Jacobides et al., 2018). It is thus neither the invisible hand of the market nor the visible hand of the hierarchy alone that guides ecosystems. Rather, orchestration—coordination without "full hierarchical fiat" (Jacobides et al., 2018)—is a combination of various coordination mechanisms and governance structures. As Dhanaraj and Parkhe (2006: 659) note, orchestration represents an "*interesting situation in which hub firms lack the authority to issue commands and autonomous network members are not obliged to obey*."

Thus, for such an ecosystem of heterogeneous organizations to function effectively, it must be adeptly orchestrated by a central actor who, for reasons we will elaborate on later, assumes a central role (Dhanaraj & Parkhe, 2006). As knowledge and value creation activities are dispersed across the ecosystem, the lead firm(s) must facilitate their flow and actively engage participants to ensure the realization of focal value (Adner, 2017; Dhanaraj & Parkhe, 2006). This engagement is contingent on the premise that these actors can capture sufficient value—greater than what they would achieve independently or within competing ecosystems (Dhanaraj & Parkhe, 2006; Foss, Schmidt, & Teece, 2023). While lead firms lack full hierarchical fiat, they can employ various strategic practices to indirectly align ecosystem actors, maintaining a delicate balance between collective value creation and individual value capture. Additionally, by building relational capital, they can navigate the trade-off between knowledge sharing and the protection of proprietary assets (Kale, Singh, & Perlmutter, 2000). In light of these notions, earlier studies have posed a pivotal question: how can a lead firm effectively orchestrate external actors in the absence of hierarchical authority (Dhanaraj & Parkhe, 2006)?

Over the years, numerous studies have contributed, both directly and indirectly, to answering this question. For instance, recent conceptual studies and reviews have advanced our understanding of ecosystem structures in view of interdependence between actors (Adner, 2017; Jacobides et al., 2018), exploring the competitive dynamics between ecosystems (Rietveld & Schilling, 2021), and outlining different layers of ecosystem orchestration (Autio, 2022). These ecosystems are prevalent across industries and fields of inquiry, attracting scholarly attention in the domains of innovation (Adner & Kapoor, 2010; Ritala, Agouridas, Assimakopoulos, & Gies, 2013; Walrave, Talmar, Podoynitsyna, Romme, & Verbong, 2018), entrepreneurship (Pitelis, 2012; Shen, Guo, & Ma, 2023), information systems (Song, Xue, Rai, & Zhang, 2018), and marketing (Hewett, Hult, Mantrala, Nim, & Pedada, 2022; Li, Claes, Kumar, & Found, 2022). In addition, two recent works were particularly relevant to advancing our understanding of ecosystem orchestration. First, a review paper by Altman et al. (2022), while focusing more on structure and not providing an extensive overview of orchestration mechanisms, offers a useful starting point by conceptualizing orchestration as a coordination approach where the locus of activity is primarily external, while the locus of control remains internal. Second, Chen et al. (2022), while focusing on digital platforms, examine control mechanisms through the lens of organizational governance and corporate strategy. Collectively, these studies clarify how orchestration differs from other coordination approaches and emphasize its complex nature, shaped by the interplay of multiple mechanisms.

Yet, surprisingly, despite more than two decades of research on ecosystems, no comprehensive review has been carried out to integrate the findings on orchestration mechanisms deployed by lead firms. On the contrary, in trying to keep pace with this ubiquitous phenomenon, research has produced an unwieldy array of ecosystem types, resulting in conceptual incoherence and literature fragmentation, and prompting some to question the very value of the concept itself. Prior studies note that some observers have even dismissed it as a convenient catch-all term rather than a meaningful analytical construct (Autio, 2022; Thomas & Autio, 2020). That being said, systematic and extensive reviews specifically focused on the mechanisms of ecosystem orchestration remain very limited, with most addressing the topic only indirectly or in passing, with vague definitions and little systematic investigation. As Table 1 shows, existing reviews remain limited in scope: they either focus on a particular ecosystem type, or stop short of examining how orchestration is actually enacted. Thus, despite the growing prominence of ecosystem orchestration and the expanding body of research on the subject, the fundamental mechanisms underlying this coordination approach have yet to be systematically reviewed or integrated.

Insert Table 1 about Here

Because the mechanisms of coordination and control are often overlooked, our understanding of how orchestration actually unfolds remains limited. As a result, ecosystems continue to appear as black boxes, leaving unresolved the question of how value is co-created and how lead firms manage to align and steer participating actors. The literature is also fragmented across innovation, digital, industrial, entrepreneurial, and platform ecosystems (Thomas & Autio, 2020), lacking a unifying framework that identifies core *orchestration mechanisms* and how these mechanisms are configured into distinct *orchestration regimes*. Against this backdrop, the purpose of this article is to provide an integrative review of ecosystem orchestration, aiming to develop a more holistic and comprehensive understanding of the mechanisms employed by lead firms. Rather than merely summarizing existing studies, this review looks under the hood to examine the underlying mechanisms of ecosystem orchestration in detail. A further objective is to bring coherence to the fragmented literature by classifying ecosystem types according to the orchestration regimes on which they depend. Guided by the central question—*How do lead firms orchestrate ecosystems without direct authority over their actors?*—we systematically select, review and analyze 226 articles on ecosystem orchestration and governance.

First, we identify that all mechanisms can be classified as codified, cognitive, incentive-based, or hierarchical. We emphasize that orchestration is fundamentally about the application of these mechanisms in combination, which varies substantially across contexts and ecosystem types. This variation and interplay largely account for the complexity, fragmentation, and perceived messiness of ecosystem theory, as opposed to a neat dichotomy of *markets and hierarchies*. Therefore, we move beyond the generic notion of orchestration and integrate the ecosystem literature by proposing a concrete typology (Cornelissen, 2017; Doty & Glick, 1994) that illustrates how orchestration mechanisms are configured into regimes shaped by an ecosystem's degree of codification and level of centralization. By emphasizing the varying degrees of codification and centralization that define ecosystem orchestration, we distinguish orchestration regimes from governance modes and other general coordination concepts. We offer a direction for ecosystem orchestration research by framing it as a dynamic configuration of mechanisms that vary and conjoin depending on context. This clearer distinction not only sharpens the conceptual boundaries of orchestration but also enables classification of ecosystem types based on their dominant orchestration mechanisms, in addition to their value logic and core purpose.

Second, by integrating findings on the coordination approaches used by lead actors, we aim to advance ecosystem theory and anchor it more firmly within classic theories of the firm. We demonstrate its connections to established theories such as the resource-based view (RBV), property rights theory (PRT), transaction cost economics (TCE), and the knowledge-based view (KBV), thereby broadening the appeal and relevance of ecosystem orchestration to a wider management audience as well as to public administration research. We argue that ecosystem scholars' engagement with a wide array of classic theories is neither coincidental nor an attempt to legitimize an inherently broad construct. Rather, it reflects the inherent complexity of new organizational forms that cannot be captured by canonical and stylized models of markets and hierarchies (Chen et al., 2022). Ecosystems differ not because they introduce principally new mechanisms, but because they often combine and apply established ones in novel, complex configurations. This is the primary reason why classic theories of the firm remain highly relevant for understanding ecosystems, yet no single theory alone can fully account for these complex configurations of mechanisms involved in such hybrid forms. By clarifying how the ecosystem concept links back to each of these classic theories of the firm and by highlighting the understudied aspects of ecosystem orchestration, we aim to move the field from a seemingly haphazard patchwork toward a more coherent and informed research agenda, as well as to challenge the view of those who still regard ecosystems as merely a catch-all term. This is crucial in an era of increasing interdependence among organizations, where lead actors seek to establish control over the activities of actors they do not own.

REVIEW METHOD AND SCOPE

This paper undertakes a comprehensive review of the ecosystem orchestration literature. While our approach seeks to uphold transparency, reproducibility, and methodological systematism (Denyer & Tranfield, 2009; Kraus, Breier, & Dasí-Rodríguez, 2020; Rousseau, 2024; Tranfield, Denyer, & Smart, 2003), at the same time, we aim to synthesize insights from a literature that has become increasingly fragmented across various subfields with different ecosystem types.

Integrative reviews are particularly well suited for overcoming disciplinary fragmentation, transcending disciplinary boundaries and consolidating findings in emerging fields that lack extensive conceptual development or an established theoretical framework (Cronin & George, 2023; Torraco, 2005). Accordingly, the purpose here is integration and synthesis through methodological systematism in article selection and analysis, moving beyond a purely descriptive review (Fan, Breslin, Callahan, & Iszatt-White, 2022; Rousseau, Manning, & Denyer, 2008).

We adopt Adner's (2017) widely used definition and view ecosystems as structures organized around a focal value proposition. Thus, to maintain conceptual coherence and remain consistent with this definition, we did not include "networks" or "platforms" as search terms. Although ecosystem orchestration has been influenced by Dhanaraj and Parkhe's (2006) earlier work on innovation networks, the two concepts have diverged over time, with ecosystems distinguished by the more central role of coordinating entities (Kinder, Six, Stenvall, & Memon, 2022) and being organized around a clear collective output (Shipilov & Gawer, 2020). Similarly, following Gawer and Cusumano (2014), we view platforms as a potential technological core of an ecosystem, but not equivalent to the ecosystem *per se*. Therefore, omitting the word "platform" from our search string does not exclude platform ecosystems but rather excludes studies that focus solely on this technological core without addressing the broader ecosystem. In addition to *orchestration*, we included *governance* in our search,[2] because the two terms are frequently used interchangeably for similar purposes in the context of ecosystems.

To refine the search, we applied proximity operators and filters (see the appendix for a detailed audit trail). Our search targeted references to ecosystem orchestration and governance within titles, abstracts, and keywords across the Scopus and Web of Science (WoS)—the two most

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[2] We thank an anonymous reviewer for this suggestion.

extensive academic databases (Kraus, Breier, Lim, Dabić, & Kumar, 2022). We have excluded non-management journals from our search, as including them would result in a sample dominated by environmental and natural science papers using the term "ecosystems" in its original, biological sense. By applying a filter for management and business journals, Scopus/WoS still includes such outlets as *MIS Quarterly*, *Public Management Review*, *Governance*, and various information systems and social science journals. All of these are listed in the 2024 CABS Academic Journal Guide, which we used to ensure journal quality (Burström, Lahti, Parida, & Wincent, 2024; Kraus et al., 2022), while retaining interdisciplinarity by keeping the initial inclusion threshold relatively low.

Throughout the selection process, we excluded articles focusing on the orchestration of policy regimes, those that mentioned orchestration only in passing or as a buzzword, as well as papers on IT orchestration (which is a different concept, related to software automation). We also excluded papers that did not adequately define an "ecosystem" or used the term synonymously with dyadic alliances, loose networks, value chains, or merely as a generic umbrella term. At this stage, we further excluded publications in which notions of orchestration were overly descriptive or lacked sufficiently rigorous evidence. At the same time, we included some additional studies that we snowballed as our review progressed. These criteria resulted in a final sample of 226 papers.

In the analysis, to extract explicit data on ecosystem orchestration, we employed a thematic analysis grid (Anderson, Lees, & Avery, 2015)—a matrix-based method in which data sources are organized by rows and questions (e.g., orchestrated by "whom," "how," "where") by columns. The corresponding cells in the data sheet were populated with our notes and verbatim quotes from the texts. As we reviewed the papers and completed the grid, we examined each cell to identify the specific coordination mechanisms (e.g., APIs, trust, IP), theoretical foundations (e.g., RBV, TCE), and contextual features (e.g., e-commerce in Asia, organic food industry in Brazil). This structured dataset formed the foundation for our synthesis, through which we identified the orchestration mechanisms and regimes discussed in the following sections.

DISTILLING ORCHESTRATION INTO DIFFERENT MECHANISMS

While recent contributions, in particular those of Adner (2017) and Jacobides et al. (2018), which are frequently cited in the studies we have reviewed, helped close important conceptual gaps and provided a more solid foundation for ecosystem research, the notion of *orchestration* remains far less developed. In much of the literature, the term is invoked without definition, used synonymously with other coordination approaches, diluting its conceptual value, or mentioned without specifying the mechanisms through which it operates. Consequently, knowledge of how ecosystems are orchestrated remains dispersed, with little integration of the mechanisms at play. Our review seeks to remedy this by distilling the vague concept of orchestration into four sets of orchestration mechanisms: *codified*, *cognitive*, *incentive*, and *hierarchical* (see Table 2 for representative examples). In reviewing the orchestration mechanisms employed by lead firms, we found that they consistently aligned with the above four overarching categories, which emerged inductively from a systematic analysis of each cell in the column of orchestration mechanisms in our analysis grid. In the following section, we examine each of these four, provide empirical examples from the literature, and argue that all orchestration mechanisms can be mapped onto these categories, which capture in finer detail how lead actors control and coordinate participants without direct ownership.

Insert Table 2 about Here

Codified Mechanisms

By codified mechanism, we refer to *an intermediary device or interface that encapsulates information and functions as a shared knowledge repository and reference point, reducing ongoing one-to-one exchanges and buffering participants from coordination burdens*. Such mechanism is particularly useful given that lead actors neither hold direct authority over ecosystem participants nor possess the coordination capacity to manage all interactions. Through codification and standardization, lead firms can pre-establish design rules that facilitate modularization and reduce coordination costs (Baldwin, 2008; Felin & Zenger, 2014; Ulrich, 1995), shaping actor behavior without the need for direct intervention. There are several manifestations of this mechanism, which typically vary in how effectively and in what level of detail they encapsulate information, but they generally serve the same function—as aids in communication, monitoring, and knowledge transfer.

Digital and technical interfaces. Perhaps the most evident codified mechanisms are information-technology tools such as digital code, databases, algorithms, and AI. Information technology has long been recognized as a catalyst for ecosystems and the broader trend of vertical disintegration (Argyres, 1999; Teece, 2018) and disintermediation (Kowalkowski, Kindström, & Carlborg, 2016; Sibanda, Ndiweni, Boulkeroua, Echchabi, & Ndlovu, 2020), by effectively automating many routine processes and thereby easing coordination and reducing the need for firms to internalize activities that previously had to be closely monitored. For example, platform ecosystem leaders such as SAP or Google develop and deploy application programming interfaces (APIs) and software development kits (SDKs), to enable and facilitate external developers to connect with their platforms (Chen et al., 2022), fostering innovation beyond their direct oversight (Schreieck, Wiesche, & Krcmar, 2021; Yoo, Henfridsson, & Lyytinen, 2010; Zeng, Yang, & Lee, 2023). More importantly, growing evidence of digitalization and servitization in manufacturing industries shows that these mechanisms are not exclusive to digital-platform leaders. They are increasingly employed by manufacturing firms such as BMW and MAN to develop proprietary platforms, digitally connect external developers, and add new product features even after final assembly of the physical product (Kapoor, Bigdeli, Schroeder, & Baines, 2022; Weiss, Wiesche, Schreieck, & Krcmar, 2022). However, the omnipresence and central role of digital technologies today make it easy to forget that codification need not imply digitalization. Manufacturers of physical products similarly rely on industrial standards and modular technologies that, in principle, fulfill the same integrative function as digital interfaces. For example, lead original-equipment manufacturers (OEMs) can leverage technical specifications to maintain control over external complementors and lock them into asset-specific investments (Jacobides, MacDuffie, & Tae, 2016). Essentially, whether digital or physical, standard interfaces serve as integration devices that reduce ambiguity among multilateral actors and in this way assist lead firms in orchestration.

Contracts and formal interfaces. Contractual and formal specifications, such as licensing arrangements and service-level agreements, are another subset of codification mechanisms that provide explicit definitions of exchange rules, actor roles, and the division of labor within ecosystems. In their study Reiter, Stonig and Frankenberger (2024) note how the relationship between lead firms and ecosystem actors relied on codified, formal agreements. In doing so, contractual interfaces help constrain opportunism in ecosystems where, despite system-level objectives, actors primarily pursue their own interests and lead firms must continually balance value creation and value capture. Thus, by formalizing the rules of engagement through contracts, the need for internalization can be reduced, giving way to ecosystem-type arrangements (Holgersson, Baldwin, Chesbrough, & Bogers, 2022). Overall, information encapsulation, whether technical (design rules) or contractual (exchange rules), assists orchestration by reducing the frequency of multilateral interactions among actors through the ex-ante codification of these rules on an external interface that serves as a reference point. The ecosystem thereby becomes more modular, as communications are channeled on an aggregate level through standardized module interfaces.

Cognitive Mechanisms

What we refer to as cognitive mechanisms are intangibles that, despite their tacit and uncodifiable nature, function as devices that ensure alignment and coherence in the behavior of ecosystem members. Specifically, these include mental models, cognitive frames, trust, narratives, collective identities, informal norms, social ties and relationships, and culture that help coalesce diverse actors around a common value proposition. These cannot be fully encapsulated in tangible artefacts but are instead more "up in the air"—difficult to perceive or articulate, yet instrumental in ecosystem orchestration. Similar to codified mechanisms, these are not present by default. They are developed over time by ecosystem leaders through efforts such as facilitating informal engagement, organizing conferences and knowledge-sharing platforms, diffusing best practices, establishing ecosystem legitimacy, and cultivating shared mental models among participants (Gastaldi & Corso, 2016; Sun, Wang, Zhang, & Kou, 2025). Studies have shown that the use of "*social coordination mechanisms*", such as hackathons, can promote ecosystem growth by advancing platform adoption (Fang, Wu, & Clough, 2021). These mechanisms are applied even by orchestrators in high-tech industries, where one might expect more reliance on codification. For example, Xiaomi leveraged cognitive mechanisms in creating the Mi Fan community and purposefully "*evoked emotions to generate a sense of affiliation*" (Shi, Liang, & Ansari, 2024: 1108). In another study, a healthcare platform leader has instead resorted to creating "*an urgency* *of action around a solution*", thereby motivating engagement around the respective platform (Cennamo, Oliveira, & Zejnilovic, 2022: 64). Vice versa, ecosystem leaders can leverage existing mental frames and align with an already established vision of the future. For instance, a study of an agricultural ecosystem in Kenya shows how a lead firm managed to turn political actors from barriers into enablers by aligning itself with the established vision outlined in national long-term roadmaps (Busch & Barkema, 2022). In essence, these studies collectively show that lead firms tend to rely, to varying degrees, on cognitive mechanisms regardless of whether the ecosystem in question is a digital platform or an innovation ecosystem, and that they can evoke different emotions among participants to align them around a shared value proposition.

As contractual arrangements cannot anticipate and account for every contingency that may arise between ecosystem members, the residual, uncodifiable gaps are often patched by such cognitive mechanisms (Aagaard & Rezac, 2022). *Trust*, in particular, is especially significant and appears repeatedly in the studies we reviewed. In their study of the European medical sector, Reypens, Lievens and Blazevic (2021) find that when codified mechanisms were lacking, trust became crucial, and lead firms built alignment between both private and public actors by facilitating gradual reciprocal adjustment toward a shared consensus. Similarly, Zeng, Tavalaei and Khan (2021) find that trust-building is vital, particularly in settings where institutions are less developed, and ecosystem actors operate in more uncertain environments characterized by informal interactions. In such cases, lead firms must maintain continuous and close communication with ecosystem actors to facilitate engagement. Thus, cognitive mechanisms support orchestration by shaping shared mental models and aligning diverse actors' behaviors toward the shared mission and value proposition. They come to the forefront particularly in situations where high uncertainty and complexity make the development and application of ex-ante codified mechanisms challenging, or when lead firms lack sufficient power to impose codified rules that would be adopted by ecosystem actors.

Incentive Mechanisms

Incentive-based mechanisms act as signals of sufficient pecuniary or nonpecuniary value to potential members to secure their engagement in the ecosystem. These are the mechanisms that "pull" participation. Lead firms, for example, can artificially design natural market dynamics by introducing price competition within their platforms (Chevalier & Goolsbee, 2003). In multisided platform ecosystems, lead firms frequently subsidize one side of the network to resolve the chicken-and-egg dilemma and reach a critical mass, triggering network effects and self-sustaining growth (Nagy, Bläse, Filser, Appenzeller, & Puumalainen, 2025; Tavalaei, Santaló, & Gawer, 2025). These examples are well-documented but not exhaustive, nor are they exclusive to platform ecosystems. Our review finds that lead firms employ a wide range of approaches to incentivize participation across different contexts.

Economic incentives. In entrepreneurial and innovation ecosystems, orchestration often takes shape through resource support and nurturing. For example, a government-affiliated organization in Singapore purposefully orchestrated an innovation ecosystem by implementing incentive-mechanisms—such as subsidies and cost-sharing—that redistributed risk, supported capability development, and facilitated entry into international markets (Ruess, Müller, & Pfotenhauer, 2023). According to Lu and Zhang (2024), ecosystem development can be facilitated when lead firms implement growth mechanisms, including venture acceleration programs and funding initiatives, to attract and support participants. Similarly, Masucci, Brusoni and Cennamo (2020) show how lead firms can incentivize complementors by designing technologies and IP strategies that expand their opportunities for value creation and capture. Thus, notably, many of these incentive-mechanisms are economic in nature and act as signals to existing or prospective members that participation within an ecosystem offers greater benefits than operating independently.

Noneconomic incentives. Furthermore, many ecosystems comprise lead actors or participants motivated by nonmarket interests—such as nonprofits, knowledge institutions, and government agencies (Olk & West, 2023; Zhou, Tang, Li, & Tong, 2025)—which means that securing their engagement requires different types of incentive-mechanisms that do not necessarily bring any direct commercial benefits (Ritala, 2024). These mechanisms feature most prominently in more recent research on ecosystems acknowledging governments and NGOs as both potential orchestrators and participants, particularly in studies informed by the circular economy and sustainable development literatures. For instance, in their study of a nonprofit-led ecosystem, Savaget, Ozcan and Pitsis (2024) showed how the lead actor delivered nonpecuniary value to a range of stakeholders, such as moral satisfaction and social legitimacy and, after making the ecosystem self-sustaining, exited without seeking to derive any ongoing commercial value. In another study of a firm-led ecosystem in the building industry, the lead actor secured the participation of knowledge institutions, NGOs, and private architects by creating highly inclusive technological standards and a shared resource pool. This approach ensured that the ecosystem's solutions delivered not only economic value to participating firms but also societal value to NGOs, research opportunities to academic actors, and technical frameworks and resources to architects, thereby engaging a highly heterogeneous set of participants through differentiated incentives (Mozheiko, Sund, & Dreyer, 2025). Broadly, economic and non-economic incentive mechanisms facilitate orchestration by drawing in and sustaining the engagement of diverse ecosystem participants and are employed equally by both profit-driven and nonprofit lead actors.

Hierarchical Mechanisms

By hierarchical mechanisms, we mean the exercise of quasi-hierarchical authority over participants, usually through ownership of nonfungible key assets that give lead actors bargaining power to influence governance structures and policies and regulate access within an ecosystem. These mechanisms "push" participation and, as the term implies, reflect a more "top-down" mode of orchestration. To be more specific, although lead actors lack ownership over external participants, they often own the platform and define its architecture and governance structure (Thomas & Ritala, 2022). Asset ownership of the digital platform and related technologies—or property rights (Chen et al., 2022)—confers power and allows them to exercise quasi-hierarchical authority by shaping gatekeeping and access policies (Zhang, Li, & Tong, 2022). This notion of power stemming from control over inclusion–exclusion decisions via ownership of critical assets is central to property rights theory (Hart & Moore, 1990). Our review suggests that this is especially pronounced in ecosystems where such power often stems from interdependence under power asymmetries inclined toward the central actor.

In digital ecosystems in particular, the lead firm typically holds substantial bargaining power over complementors (Zhu & Liu, 2018), as complementors' outputs converge on a single platform controlled by one actor. However, this leverage and ability to exercise quasi-hierarchical authority is not limited to digital ecosystem leaders alone. For example, in the aerospace industry, lead firms leverage patent ownership to control and coordinate their innovation ecosystems, balancing value creation and value capture among participants (Azzam, Ayerbe, & Dang, 2017). Similarly, in another case study, a B2B platform leader in the event management industry exercised control over platform access through its ownership rights, ensuring the quality of offerings across participants—something difficult to achieve without the ability of direct intervention into the platform's network effects (Laczko, Hullova, Needham, Rossiter, & Battisti, 2019). Thus, although lacking ownership over external participants, through ownership of the ecosystem's foundational asset (usually technology), the firm can gain substantial quasi-hierarchical control over complementors whose value creation and capture opportunities depend on access to this asset. Consequently, hierarchical mechanisms assist lead firms in orchestration by enabling them to leverage power and unilaterally set conditions for participation, thereby compelling ecosystem participants to adjust their behavior accordingly. The potential extent of this power is evident in the widespread antitrust scrutiny and regulations faced by leading platform owners (Edelman, 2015; He, College, Reimers, & Shiller, 2022).

FROM ORCHESTRATION MECHANISMS TO ORCHESTRATION REGIMES

Having classified the underlying mechanisms of ecosystem orchestration, we now move beyond classification and, informed by the extant literature, develop a typology of orchestration regimes, that seeks to balance parsimony with explanatory power (Delbridge & Fiss, 2013; Doty & Glick, 1994). While understanding the underlying orchestration mechanisms is important, examining them in isolation is insufficient. We need to understand how they are combined and reconfigured depending on contextual factors. Indeed, our review of the literature indicates that one reason the ecosystem concept is difficult to delineate—and its boundaries remain fuzzy—is that ecosystems are never orchestrated by a single mechanism but rather by multiple, often overlapping ones. However, their relative use varies significantly, allowing us to identify four ideal types of orchestration regimes. Such a typology is a "*parsimonious framework for describing complex organizational forms*" (Doty & Glick, 1994: 230), and is *"developed with respect to a specified organizational outcome"* (Doty & Glick, 1994: 232). In our case, this outcome—consistent with our definition of orchestration as the strategic application of control and coordination mechanisms without ownership—is the integration of effort across organizational boundaries to realize the focal value proposition. Accordingly, we define orchestration regimes as combinations of mechanisms strategically deployed to achieve this focal value proposition. The four ideal types of regimes are *architectural leverage*, *modular brokering*, *administrative control*, and *consensus alignment* (summarized in Table 3).

Insert Table 3 about Here

We note that the orchestration mechanisms available to lead actors are highly dependent on the degrees of codification and centralization. These two dimensions emerged inductively from our review and are also well grounded in classic organizational literatures, which we discuss later. By codification, we refer to the articulation and transfer of information or knowledge into one of the codified mechanisms described earlier, making it easily accessible to the multilateral set of partners in an ecosystem through a shared interface rather than through ongoing bilateral communications. This process enhances the modularity of the ecosystem by reducing coordination complexity, as communication is channeled at an aggregate level through module interfaces. As such, the greater the degree of codification, the stronger the reliance on codified mechanisms relative to cognitive mechanisms when orchestrating the ecosystem. By centralization, we mean the concentration of decision-making authority and control in a limited set of actors—typically a single lead firm—arising from their distinctive resources, advantageous network positions, or ownership of nonfungible assets. Importantly, this does not imply the consolidation of activities—which would correspond to vertical integration—but rather that coordination decisions and the "rules" governing value creation are determined by central actors. Accordingly, higher degrees of centralization correspond to a stronger reliance on hierarchical, as opposed to incentive- or market-based, mechanisms in the orchestration of the ecosystem. Our four types, which vary along these two dimensions, integrate both the full range of available mechanisms and the ways in which they are bundled according to various factors. In the following sections, we elaborate on each of these regimes in turn, link them to corresponding ecosystem types identified in the literature, and clarify the circumstances under which each may be preferable for ecosystem orchestration.

Architectural Leverage

Architectural leverage is a regime that exhibits high degrees of centralization and high degree of codification. Here, an orchestrator primarily relies on the combined application of codified and hierarchical mechanisms. It is perhaps appearing the most frequently in the literature because this regime is most common to platform ecosystems such as those of Amazon or Google, which have been extensively discussed in prior studies. Indeed, originally the term itself was introduced in connection to platforms ecosystems specifically (Thomas, Autio, & Gann, 2014). We retain the term in this article, because it implies that through the control of key resources or technology, the orchestrator can leverage them over the entire architecture of the ecosystem (Chen et al., 2022). The orchestrator also does so through codified mechanisms, which are captured in the term, because architecture implies deliberate design and standardization between components. Thus, we understand architectural leverage here as the unilateral design of codified mechanisms leveraged by a lead firm to control and coordinate complementary activities of ecosystem actors.

A prime example is Apple and comparable platform-based ecosystems that develop and use APIs or other codified interfaces to coordinate external actors while retaining the authority to unilaterally set rules for participation (Thomas et al., 2014; Wang, Ma, Zhang, & Li, 2024; Zhang et al., 2022). Once power becomes centralized, pricing strategies can be unilaterally determined by the platform owner with little resistance, or the platform owner may begin competing directly with its own complementors by exploiting information asymmetries (Zhu & Liu, 2018). Such “unilateral moves” (Kude & Huber, 2025) may further include restricting access or abruptly discontinuing critical APIs, depriving dependent complementors of essential resources (Cutolo & Kenney, 2021).

In addition, this regime is characteristic of industrial ecosystems, where interdependent actors coordinate their efforts through standardized components but operate within a quasi-hierarchical structure orchestrated by a central original equipment manufacturer (OEM). For example, Jacobides et al. (2016) studied how, in contrast to the computer hardware industry—where modularity and outsourcing eroded OEMs’ power and deverticalized the industry architecture—in the automotive industry, OEMs retained hierarchical control over proprietary component designs. This control enabled them to sustain architectural leverage across the broader automotive ecosystem, even as production was outsourced to complementors. While digital platform and non-digital industrial ecosystems differ primarily in the technologies underpinning their core platforms, both rely on codified and hierarchical coordination mechanisms. Thus, they differ technologically but not conceptually—a distinction that helps clarify existing confusion and reconcile fragmented strands of the literature. By developing and controlling these APIs or proprietary standards and centralizing the ecosystem’s dependence on them, lead firms enable integration on their own terms, whether to maintain the quality of ecosystem offerings or to capture greater value for themselves.

Modular Brokering

Modular brokering refers to a regime characterized by a high degree of codification but a low degree of centralization. In the absence of significant influence over complementors, the orchestrator must instead rely on economic mechanisms to engage with and coordinate ecosystem participants. A relevant example is Netflix, which operates as a platform owner. However, because many of its complementors are large and resourceful movie studios over which it has limited leverage, Netflix relies on economic incentives and revenue-sharing schemes to attract them (Nam, Ro, & Jung, 2023). In this sense, the lead actor functions as a broker among different "module providers" to assemble the ecosystem. Additionally, when developing an ecosystem that spans industry boundaries, codification often requires accessing and integrating the knowledge and capabilities of other industry leaders, which can lead to close collaboration among several lead firms in jointly building an ecosystem. For instance, in Chongqing, China, an internet platform firm partnered with leading companies in the electronics and pharmaceutical industries to pool resources for the development of a cross-industry platform ecosystem (Wang et al., 2024). Furthermore, under conditions of high codification and low centralization, multihoming often emerges—where several competing platforms coexist but hold limited bargaining power over complementors, which can also participate in rival platforms. Since multihoming may intensify competition (Pujadas, Valderrama, & Venters, 2024), lead firms are compelled to rely more on economic-incentive mechanisms rather than hierarchical ones. Consequently, when a firm is one of several ecosystem leaders or lacks architectural knowledge, it cannot wield architectural leverage, leading to a less centralized regime of modular brokering.

Our review also suggests that both modular brokering and architectural leverage regimes are most effective when the emphasis lies on exploitation rather than exploration, and when uncertainty is low. For instance, Reiter et al. (2024), in their study of ecosystems in the banking sector, highlight that contractual mechanisms tend to prevail over cognitive ones under low uncertainty, when the objective is to achieve alignment around specific issues. This is likely because, in such contexts, specification is easier than in situations of high uncertainty, where it is impossible to anticipate and define all contingencies ex ante. When the goal of both the lead firm and its complementors is not innovation but value capture—seeking clear frameworks, resources, and favorable conditions that such platforms provide—we can expect a higher degree of codification. These regimes therefore appear more characteristic of platforms and industrial ecosystems, while less applicable to innovation and entrepreneurial ecosystems, which tend to rely more on administrative control or consensus alignment, which we turn to next.

Administrative Control

What we refer to as administrative control is a regime where codification is low but centralization is high. Here, a small number of lead actors hold concentrated decision-making authority, yet must rely heavily on cognitive mechanisms to build alignment across the ecosystem. For example, in some entrepreneurial ecosystems, venture capital firms function as "active hubs," occupying a central position and exerting considerable influence over complementors. However, they attract and coordinate participants primarily by fostering trust and facilitating the sharing and integration of complementary knowledge, with limited reliance on codified mechanisms (Burström, Lahti, Parida, Wartiovaara, & Wincent, 2023). Similarly, Cobben, Neessen, Rus and Roijakkers (2023) examine a case where a family firm orchestrated an open innovation ecosystem in the healthcare sector involving both commercial and noncommercial actors. In the early stages, the firm operated as a clear central actor while relying primarily on cognitive mechanisms. Creating trust and a psychologically safe environment was vital for fostering the integration and exchange of dispersed ideas and knowledge. Beyond innovation and entrepreneurial ecosystems, this regime also appears common in public actor–led ecosystems where authority is consolidated but the orchestrator aims to cultivate a supportive environment among diverse participants. For example, Masucci, Camerani and Corrocher (2024) examine how a European business university assumed such a role by creating a university-based accelerator that orchestrated innovation processes internally—across student initiatives and departments—and externally, by leveraging complementarities with STEM universities that provided essential technical knowledge, as well as with startups and government agencies. In doing so, the university relied primarily on cognitive mechanisms such as consensus building and creating a "sense of urgency" and "sense of overall belonging," while simultaneously maintaining central control over entry into the accelerator program.

Our review further shows that administrative control appears less frequently in the literature than other orchestration regimes. A key reason why actors can centralize authority without owning external actors is their control over a proprietary, codified technical standard or digital interface. This suggests that, in ecosystem settings, hierarchical mechanisms often operate in tandem with codified mechanisms, which helps explain why administrative control regime emerges infrequently.

Consensus Alignment

Lastly, in ecosystems where both codification and centralization are low, orchestration rests on "consensus alignment" and relies primarily on a combination of cognitive mechanisms and economic/non-economic incentives. Without the ability to codify knowledge or exert bargaining power over complementors, ecosystem leaders must employ these softer levers and orchestrate the ecosystem with a gentle touch. Some earlier theoretical work in organizational studies has in fact outlined consensus-based decision making, wherein solutions to problems are pursued through cognitive and heuristic search (Nickerson & Zenger, 2004). When problems are highly complex and not decomposable into ex ante design choices that would directly guide the search for solutions, addressing them typically requires intensive exchange of dispersed specialized knowledge among multiple actors and the use of abstract cognitive maps to guide that search. As Nickerson and Zenger (2004: 626) argue: "*consensus-based hierarchy is a potential solution to the failure of authority in governing heuristic search as problems increase in complexity*."

Our review identifies a similar link between problem complexity and the use of consensus-based modes in ecosystem orchestration. We find that reliance on cognitive and incentive-mechanisms as the primary coordination tools most often appears in ecosystem studies focused on innovation and entrepreneurship, where the objects of exchange are most commonly knowledge and capabilities. In innovation ecosystems, actors concentrate on exploration and the discovery of new solutions to complex problems, rather than addressing moderately complex tasks or exploiting existing opportunities—patterns more typical of platform and industrial ecosystems. Accordingly, when the integration of distributed external expertise is pursued for innovation, excessive codification or direct authority can become counterproductive, and consensus alignment emerges as the orchestration regime best suited to governing the ecosystem.

For example, this approach is common in ecosystems where the central actor is an incubator that not only provides funding and resources but also seeks to develop a shared vision, "*going beyond the formal governance structures*" (Hernández-Chea, Mahdad, Minh, & Hjortsø, 2021: 8) through regular meetings and direct interactions to build a "*common understanding of purpose, aligned objectives, and effective and visionary leadership*" (Hernández-Chea et al., 2021: 12). Similarly, in another study of an innovation ecosystem dedicated to sharing knowledge to address the highly complex challenge of Alzheimer's disease, governance was extremely distributed and rested on consensus alignment across national and international government agencies, large and small pharmaceutical firms, and R&D consortia, among others (Olk & West, 2023).

This regime also appears common in innovation ecosystems pursuing a more circular economy, where integration often requires a shift in mindset. In the agricultural sector, Zucchella and Previtali (2019) show that actors in such ecosystems largely depend on informal, trust-based mechanisms and free information sharing. While a central actor initiates the process to build a shared vision and foster alignment, they rely less on hierarchical mechanisms and more on incentive-mechanisms—namely, the substantial cost savings generated by developing an innovative, lower-cost waste-recycling solution.

Finally, public management articles within our review reveal comparable regimes in public actor-led and regional ecosystems, where orchestration is framed as "collaborative governance" and is grounded in strong trust relations (Kinder et al., 2022). In such settings, change is enacted not by a single dominant entity but by collective intent and distributed leadership across participants (Kinder, Stenvall, Six, & Memon, 2021). Overall, in contrast to regimes characterized by high codification and high centralization, regimes of low codification and low centralization primarily arise in situations where integration of effort is oriented toward exploration and thus appear most commonly in innovation and entrepreneurial ecosystems. It is also notable that these ecosystems are usually regional—tied to a particular local context—which is unsurprising given their reliance on close ties and ongoing interactions. Finally, it is not uncommon for such ecosystems to have a government entity or public organization as a central actor, seeking to foster a conducive environment in the regional ecosystem and drive broader systemic change.

Ecosystem Development and Regime Changes

Ecosystems are not fixed in default, static governance modes but are dynamic—undergoing orchestration regime change as they mature. These changes reflect variations in ecosystem size, degrees of centralization and codification, uncertainty in the solution space, and the shifting emphasis between value creation and value capture. Figure 1 illustrates our typology as a heuristic tool, mapping how regimes vary along dimensions of codification and centralization. As noted earlier, these are ideal types and therefore highly stylized. In practice, lead actors may deploy several or all of the mechanisms to some degree. Nevertheless, their relative use does vary substantially and shifts as ecosystems evolve.

Insert Figure 1 about Here

Although interactions are typically loose in the early stages, and it is impossible to codify agreements ex ante in formal contracts, codification often increases as uncertainty diminishes, roles and functions become more structured, and a clearer architecture emerges as ecosystems mature (Furr & Shipilov, 2019). While early-stage activity tends to focus on value creation and innovation, attention often shifts toward value capture and exploitation over time. Others contend that when processes move from scientific and innovation-driven efforts toward commercially oriented goals, tacit knowledge tends to give way to more codified forms of knowledge (Hurmelinna-Laukkanen, Möller, & Nätti, 2022). Indeed, empirical work provides support for this pattern. In their process study of AstraZeneca's ecosystem, Remneland Wikhamn and Styhre (2023) demonstrate how the lead firm initially prioritized value creation in the ecosystem's formative phase. As the ecosystem entered the "integration" stage, this emphasis pivoted toward value capture. To enable this transition, AstraZeneca established a dedicated cross-disciplinary team with sufficient seniority and authority, specifically dedicated to harnessing value emerging from the hub and translating ecosystem-level value creation into organizational value capture (Remneland Wikhamn & Styhre, 2023). Accordingly, as the focus moved from exploration to exploitation, a more centralized structure emerged.

Furthermore, trust-based and cognitive mechanisms function more effectively when the number of actors is relatively small. As ecosystems grow, the expanding actor base makes informal coordination insufficient, and a more structured and centralized orchestration becomes increasingly necessary. Cobben and colleagues (2023) illustrate this general dynamic in their retrospective study of ecosystem change, showing that continual expansion eventually creates a need for governance arrangements that reach beyond informal, trust-based interactions to remain viable over time. Such patterns of increasing standardization and more selective member inclusion have been found, for instance, in industrial ecosystems undergoing circular business model transformations (Parida, Burström, Visnjic, & Wincent, 2019). With more members, more centralized control tends to emerge, as it becomes increasingly difficult to negotiate shared consensus among numerous actors and as network opacity rises (i.e., difficulty in tracing relationships between actors) (Reypens et al., 2021). A more hierarchical mode of orchestration can directly assign and partition responsibilities across ecosystem actors, shaping relationships by design rather than in organic manner. In addition, as an increasing number of participants converge on a central platform, the bargaining power of the platform leader consolidates.

Importantly, changes in codification and centralization are not merely emergent but often deliberate and strategic, as firms take purposive steps seeking to reshape how an ecosystem is orchestrated. For example, by introducing a technological solution that enhances value creation, a complementor can eventually move into a more central role and even become an orchestrator

(Miehé, Palmié, & Oghazi, 2023). Thus, centralization may shift not only in degree but also in direction—toward a specific actor. Furthermore, in more decentralized ecosystems without a single lead firm or pronounced power asymmetries, centralization can also emerge through the formation of strategic alliances that constitute the ecosystem's core and act as a collective governing body (Jacobides et al., 2018; Shipilov & Gawer, 2020). As Olk and West (2023) show, the absence of a coordinating entity can lead to inefficiencies and duplicated efforts, prompting a shift toward centralizing certain decisions within consortia. Although governance in their case remained distributed, some structure was introduced by centralizing and clearly articulating specific decisions to prevent resources from being spread too thin. Their findings indicate that although consortium members were heavily involved in the technical aspects, they played a far more limited role in administrative tasks, which were instead managed by a centralized consortium staff. In another example, Mozheiko et al. (2025) explore how a lead actor took strategic steps to orchestrate an ecosystem by first uniting participants in the building industry around a shared vision (cognitive mechanisms). As the ecosystem expanded, this vision was subsequently translated into a set of technical industry standards, tools, and policies that codified its key parameters while also establishing a formal alliance that acted as the governing body to consolidate resources, coordinate multilateral actors, and prevent overlaps. Thus, even in the absence of a dominant firm, coordination often becomes consolidated within a single governing entity—typically an alliance or consortium formed by ecosystem members. When several actors share leadership responsibilities, centralizing coordination in this manner helps prevent redundant efforts. Overall, the general pattern is that as the dust settles and uncertainty is dispelled, a more structured, centralized ecosystem architecture tends to materialize.

That being said, there can also be exceptions in which the process moves in the opposite direction, particularly when attention shifts back toward more exploratory forms of innovation. For example, Mazzucato and Robinson (2018) explain that NASA's earlier "mission-oriented objectives" enabled it to direct and tightly govern the activities of ecosystem participants, whereas the later emergence of a broader innovation ecosystem with more "open-ended objectives" shifted NASA's role toward facilitating and funding diverse complementary actors. A move toward consensus-oriented coordination may likewise emerge as network relationships become more transparent and less opaque (Reypens et al., 2021). In some cases, a lead firm may pursue a short-lived nonmarket objective through ecosystem orchestration. Once this objective has been achieved, the firm may reduce its participation, diminishing centralization and creating a governance void that forces the ecosystem to restructure itself (Mozheiko et al., 2025). Ultimately, the purpose behind integrating dispersed knowledge and activities of ecosystem actors—whether oriented toward exploitation or exploration, or toward nonmarket or market strategies—guides the selection of orchestration mechanisms how they are combined into distinct orchestration regimes. As this purpose shifts, corresponding changes in centralization and codification may follow.

LINKING ECOSYSTEM ORCHESTRATION LITERATURE TO CLASSIC THEORIES OF THE FIRM

Our literature review clearly indicates that the field remains highly fragmented and is only beginning to develop a coherent foundation. The inherent complexity of the domain is unsurprising, as it concerns an open system involving multiple actors and diverse value propositions, with boundaries that are inherently fuzzy and inevitably—somewhat arbitrary. This arbitrariness, together with the absence of a shared foundation, is reflected in the plurality of theoretical perspectives and literature streams on which ecosystem studies rely, as well as in the adjacent phenomena they seek to inform. For instance, scholars have examined ecosystem orchestration in relation to open innovation (Gutmann, Chochoiek, & Chesbrough, 2023; Remneland Wikhamn & Styhre, 2023), the circular economy (Blackburn, Ritala, & Keränen, 2023; Langley, Rosca, Angelopoulos, Kamminga, & Hooijer, 2023), and business models (Gomes, Farago, Facin, Flechas, & Silva, 2023; Hou, Cui, & Shi, 2020; Mozheiko, 2025), among others. Although cross-fertilization across domains and theoretical boundaries can be fruitful, the borrowing of terms and theories from disparate fields often reflects convenience rather than a genuine attempt to establish meaningful connections. The notion that ecosystem research is approached from virtually any angle and attracts scholars across diverse domains highlights not only the wide-ranging appeal of the concept but also its complexity and lack of a coherent foundation.

However, through our review, by examining the mechanisms through which orchestrators integrate effort across organizational boundaries, we identified several meaningful connections to classic theories of the firm. Articulating these connections can help establish greater coherence by anchoring the ecosystem unit of analysis more firmly in established theories. In particular, we find that ecosystem scholars sometimes draw explicitly on the resource-based view (Zeng et al., 2023), the knowledge-based view (Zámborský, Ingršt, & Bhandari, 2023), transaction cost economics (Foss et al., 2023), and property rights theory (Cunningham, Menter, & Wirsching, 2019), but more often do so implicitly by discussing the very mechanisms we identified in our review that clearly reflect these theoretical perspectives. Prior work indicates that the various theories of the firm are often interrelated (Argyres, Felin, Foss, & Zenger, 2012; Gibbons, 2005). Since ecosystems represent a higher and more complex level of analysis than individual firms or bilateral alliances, a holistic understanding of ecosystem orchestration demands integrating multiple theoretical perspectives. It is therefore unsurprising that scholars draw on a range of diverse theories, as no single one is sufficient to capture the inherent complexity of ecosystem orchestration.

To advance our understanding of the inherent connections between ecosystems and classic theories, we now turn to examine how traditional theories of the firm—knowingly or unknowingly—inform ecosystem research, and how findings from the latter can, in turn, refine these theories that have traditionally focused on the dichotomy between markets and hierarchies. By establishing connections with the RBV, PRT, TCE and KBV, we demonstrate that the ecosystem lens—and the associated use of the concept—is often theoretically justified. The field's fragmentation and breadth reflect precisely the fact that ecosystems are orchestrated through combinations of mechanisms emphasized across these different theories, while none of which alone can fully capture this complexity. In doing so, we aim to place ecosystem research on track by raising ecosystem scholars' awareness of theoretical connections that are often only implicit.

Resource and Capability Theories

When examining the theoretical underpinnings of the ecosystem orchestration literature, the Resource-Based View (RBV), as well as the subsequent capability theories, stand out as the most prominent frameworks. Seminal writings by Barney (1991; 2001) posit that sustainable competitive advantage stems from a firm's internal bundle of valuable, rare, inimitable, and non-substitutable (VRIN) resources. In a similar vein, Teece and colleagues (1997; 2007) advance that sustained competitive advantage also rests on firms' ability to hone internal managerial, technological, and organizational processes, as articulated in the dynamic capabilities framework. Modern ecosystem literature frequently draws on these foundational writings, increasingly recognizing that the value creation potential of an ecosystem hinges on the unique resources and capabilities contributed by its participants (e.g., Adner, 2017; Teece, 2018), and that orchestrating an ecosystem of partners serves as an important vehicle for firms to gain competitive advantage. Thus, competition becomes increasingly characterized by ecosystem-level, rather than just firm-level considerations, and by the organization of external resources alongside internal ones (Ansari, Garud & Kumaraswamy; 2016; Moore, 1996).

A particular stream of the RBV turns to resource orchestration by shifting the analytical lens toward managerial actions (Sirmon, Hitt, Ireland & Gilbert, 2011; Sund, Barnes & Mattsson, 2018). This stream focuses on what firms do with their resources, such as how they structure their resource portfolios, bundle resources into capabilities, and leverage those capabilities to create and capture value. While the ecosystem orchestration and resource orchestration literature streams have emerged independently from each other—with the latter grounded in the resource-based view—several studies in our sample, seemingly unaware of this distinction, apply the resource orchestration lens to ecosystem orchestration (e.g., Wang et al., 2024; Zeng et al., 2021). In doing so, they implicitly suggest to us that ecosystem orchestration can be viewed as a form of external resource orchestration, offering a basis for linking ecosystems and RBV. Such connections between capabilities, resources, and ecosystem orchestration imply that effective strategic management in modern firms increasingly centers on implementing appropriate orchestration regimes to integrate resources across firm boundaries and on developing the dynamic capabilities required to reconfigure these regimes as ecosystems evolve. Yet, links to established theories of resources and capabilities remain very implicit and weakly articulated in the extant ecosystem orchestration literature.

As a form of quasi-hierarchical control, orchestration in ecosystems offers the opportunity to capture value not just from owned resources at the firm level, but also from resources at the level of ecosystems that reside outside the boundaries of any single firm (e.g., specialized complementary assets, unique technologies, market access, user communities) (e.g., Shi et al., 2024). In other words, the ecosystem orchestrator does not necessarily own the resources but can control or influence access to them and their integration. The competitive advantage thus shifts from solely relying on internal VRIN resources to excelling at externally organizing VRIN resources of others in the ecosystems (Helfat & Raubitschek, 2018; Teece, 2018). In this sense, the orchestrator's capability to efficiently and effectively identify, attract, integrate, and leverage valuable, rare, and hard-to-imitate resources owned by others within a coherent ecosystem becomes foundational to achieve competitive advantage.

In sum, resource and capability theories offer an important foundation for theorizing ecosystem orchestration, but their firm-centric legacy must be broadened to encompass the integration of externally dispersed resources. In this sense, the ecosystem orchestration literature redefines competitive advantage as an outcome of managing both internal and external resources in interconnected environments. At the same time, the theoretical traditions of resource and capability theories remain highly significant for advancing future research on ecosystem orchestration. As captured by our typology, a focal firm must deploy appropriate orchestration regimes in order to access, combine, and integrate resources and capabilities that are dispersed outside its boundaries.

Property Rights Theory

A theoretical tradition that is explicitly concerned with the granting and restriction of access is property rights theory (PRT). It explains how ownership of an asset confers residual control rights—that is, the authority to make decisions in contingencies not fully specified by contract ex ante (Grossman & Hart, 1986; Hart & Moore, 1990). Because owners possess the rights to exclude, allocate access, and to decide how an asset may be used, PRT highlights the central role of asset ownership in shaping incentive alignment, bargaining power, and organizational coordination (Chen et al., 2022; He, Tong, & Xu, 2022). When applied to ecosystems, PRT suggests that the hub firm's ownership of the core technological or architectural asset constitutes the primary foundation enabling orchestration. In such settings, the core asset—whether a digital platform, an API specification, a technology architecture, or a proprietary algorithm—anchors the flows of knowledge, value, and coordination. Ownership of this asset grants the hub firm not only authority over ecosystem evolution but also the power to shape participants' activities and engagement within the ecosystem (Boudreau, 2017; Zhang et al., 2022).

Given these notions, PRT most directly maps onto incentive-based and hierarchic orchestration mechanisms. Incentive design is fundamentally concerned with the allocation of rights and payoffs associated with access to an asset. In ecosystems, participants engage in exchange and coordination only when the expected benefits exceed their outside options. Because access to the core asset is indispensable for value capture, the hub firm, by virtue of ownership, controls the economic architecture of participation. Ownership confers upon the hub firm the power to define revenue-sharing formulas, fee structures, ranking rules, data-access privileges, and other payoff allocations that shape complementors' incentives (Chen et al., 2022; Zhang et al., 2022). PRT therefore explains why incentive-based mechanisms are the orchestrator's most consequential tool: they operate directly through the owner's residual rights to structure access and distribute the surplus generated by the ecosystem.

Furthermore, while the hub firm lacks conventional hierarchical fiat, ownership of core assets grants it a form of quasi-hierarchical authority: it can impose decisions, enforce compliance, govern inclusion and exclusion, and intervene selectively in the face of misalignment (Chen et al., 2022). In other words, asset ownership effectively creates a hierarchy-like power asymmetry, because the hub firm can unilaterally withdraw access, tighten standards, modify interfaces, or redesign the system architecture. In PRT terms, this reflects the owner's right to exclude—one of the most critical forms of residual control (Hart & Moore, 1990). Complementors comply not because they are employees or subordinates, but because their ability to create and capture value depends on continued access to the hub firm's asset. The threat of exclusion, though rarely exercised, gives hierarchical orchestration its force (e.g., Azzam et al., 2017; Zhang et al., 2022). From a PRT perspective, hierarchical mechanisms thus represent the hard edge of ownership-based authority, the point at which residual control rights substitute for formal hierarchy.

Taken together, PRT helps explain why not all ecosystem participants become orchestrators and why the power to design incentives, enforce access conditions, and codify system architecture is concentrated in the hub firm. From a PRT perspective, orchestration is fundamentally a matter of who owns the asset on which others critically depend. Ownership grants the hub firm the right to define incentives, enforce quasi-hierarchical control, and embed rules in codified artefacts; taken together, these mechanisms enable the hub to coordinate without full authority and to integrate distributed effort at scale. This asset-based foundation distinguishes ecosystem orchestration from looser forms of network coordination and from purely relational governance.

As such, PRT lens also carries important implications for future ecosystem research. It highlights the need to examine how different ownership regimes (e.g., shared ownership, open-source foundations, open-core models) (see Zhou et al., 2025) may alter the hub's ability to deploy incentive-based and hierarchical mechanisms. It invites new work on the strategic use of codification as an instrument of structural power, and it suggests that incentive design should take a more central place in theories of ecosystem orchestration and evolution. Most importantly, it recasts orchestration not as a mysterious form of "soft power," but as a systematic and theoretically grounded exercise of residual control rights by the owner of critical core assets.

Transaction Cost Economics

Transaction-cost economics is another prominent lens through which ecosystem orchestration has been examined. Rooted in Williamson's (1975, 1981) exchange-efficiency perspective on firm vertical scope (Zhang & Tong, 2021), TCE frames the organization of inter-firm exchanges as a dichotomous choice between markets and hierarchies, based on a comparison of the costs of internalizing versus externalizing transactions (Williamson, 1975, 1985). However, since "*business ecosystems surround, permeate, and reshape markets and hierarchies*" (Moore, 1996: 31), ecosystem orchestration offers an alternative to the hierarchy–market divide and points to ecosystems as hybrid structures (Chen et al., 2022; Kretschmer, Leiponen, Schilling & Vasudeva, 2022). Accordingly, orchestrating in ecosystems extends the traditional market–hierarchy dichotomy central to the transaction cost economics (TCE) approach into a broader continuum. Internally, ecosystem orchestration lowers transaction costs during firms' search processes (Pitelis, 2012; Wen et al., 2022) and reshapes collaboration behaviors (Schilling, 2015). Externally, it enables relation-specific investments by partners, thereby reducing transaction costs by mitigating the risk of hold-up (Biggar & Heimler, 2021; Jacobides et al., 2018). By enacting a quasi-hierarchical approach, focal firms orchestrate other ecosystem participants to create value collaboratively in ways that transcends a simple dichotomy between price signals and authority-based control. As such, orchestration needs to be understood as a distinct hybrid mode that can offer a more flexible configuration with lower internal costs than hierarchies (Wen, Forman, & Jarvenpaa, 2022), while at the same time incurring lower exchange costs than markets (Holgersson et al., 2022; Jacobides & Winter, 2005).

In particular, transaction cost economics can be linked to the codified and hierarchical mechanisms of ecosystem orchestration that we outlined in our review. By identifying the relationship-specific assets (Shelanski & Klein, 1995; Williamson, 1975), orchestration in an ecosystem features the contractual specificity as a form of governance mechanism, much like in other inter-firm arrangements such as alliances or supply chains. Yet ecosystem research departs from these dyadic arrangements by advancing a meta-organizational perspective (Gulati, Puranam & Tushman, 2012) that emphasizes multilateralism (Adner, 2017; Davis, 2016; Shipilov & Gawer, 2020), thereby "*presenting a different canon*" (Jacobides et al., 2018: 2265). In this sense, ecosystem orchestration goes beyond traditional dyadic, inter-firm transactions (Jacobides et al., 2018), which are typically governed primarily through formal contractual mechanisms. Instead, it encompasses triadic interactions bound by informal contracts and often characterized by high levels of technological codification, such as APIs (Teece, 2018). If low codifiability (e.g., the difficulty of specifying complete contracts) leads to greater integration and reliance on hierarchical governance, as TCE suggests, so too should this logic offer a partial explanation for the opposite trend, namely the increasing prevalence of inter-firm arrangements, given that codifiability has become increasingly possible through new technologies. With such high codifiability of not only contractual arrangements but also technological specifications, focal firms can build trust and credibility among ecosystem partners even in the absence of formal contractual bindings typical of traditional inter-firm arrangements (Shi, Liang, & Luo, 2023). This, in turn, helps achieve integration of effort *across* dispersed actors while lowering transaction costs *within* ecosystems.

However, while transaction-cost lens is well suited to analyzing inter-firm transactions, it conceptualizes technology as something hardwired and static (Baldwin, 2008). This reflects the historical context in which TCE developed, when communication and information technologies played a far more limited role in coordinating exchanges across firm boundaries. With the growing diffusion of codified digital mechanisms and firms' increasing capacity to strategically develop proprietary software and APIs, contemporary applications of TCE must explicitly incorporate technology as an endogenous resource rather than treating it as a static background condition. This reconceptualization is important because, in today's "ecosystem economy", codification is expressed not only through contractual rules of exchange but also through technological rules of design.

Knowledge-based View

In contrast to TCE, the knowledge-based view (KBV) stands out as a theory of the firm that explicitly treats codification as endogenous (c.f. Håkanson, 2010). Yet surprisingly, given the salient role of codified and cognitive mechanisms, KBV remains largely absent from ecosystem orchestration research. Whereas, in particular, the RBV and TCE recur frequently in the reviewed literature, references to KBV are rare and usually cursory. Although this theory is surprisingly rarely invoked explicitly in the ecosystem studies we reviewed, knowledge as an object of orchestration appears regularly—particularly in innovation ecosystem research, where integration typically aims at generating emergent outcomes through knowledge transfer and the joint learning of multiple actors. Despite this clear conceptual connection, the relationship remains understudied: insights from the KBV have yet to meaningfully inform ecosystem scholarship, and ecosystem research has not been actively used to advance the KBV. Here, we seek to address this shortcoming by clarifying how the ecosystem phenomenon relates to—and can be meaningfully understood through—the knowledge-based view.

KBV scholars conceptualize firms as "institutions for integrating knowledge" (Grant, 1996: 109), that outperform markets in transferring tacit knowledge among individuals—particularly when these individuals come from different "epistemic backgrounds" with varying cognitive frames and mental models (Håkanson, 2010; Kogut & Zander, 1992). Efficient integration requires establishing some degree of commonality and shared cognitive schemas, which also makes the process slow and costly due to the reciprocal nature of interdependencies in consensus-based decision making and the challenges of transferring tacit knowledge (Grant, 1996). As Kogut and Zander (1992) emphasize, firm growth depends on establishing coherent organizational logics and shared interpretive frameworks that allow many individuals with different roles and aims to work in a unified manner. Accordingly, the division and subsequent transfer of knowledge are closely tied to economies of specialization but also to rising coordination challenges (Becker & Murphy, 1992). These observations apply equally—and arguably even more strongly—to ecosystems comprising numerous specialized firms that, as we find, rely heavily on cognitive mechanisms to transfer and integrate knowledge *across firm boundaries* while also exhibiting a tendency to mitigate coordination costs through codification practices.

However, making tacit knowledge more transferable through codification is not always feasible (Balconi, Pozzali, & Viale, 2007; Spender, 1996). High task complexity and substantial uncertainty typically necessitate greater use of intensive, non-standardized coordination (Grant, 1996). Indeed, as we find, in innovation ecosystems oriented toward the exploration of the "unknown," knowledge is highly complex, requiring comparatively greater reliance on cognitive mechanisms to shape shared understandings, and longer-term, non-transactional relationships to reach consensus. By contrast, ecosystems oriented toward exploiting the "known," where uncertainty around the problem has been reduced to a requisite degree, tend to rely more frequently on codified and hierarchical mechanisms. This mirrors core insights from the KBV, namely that consensus-oriented coordination is particularly valuable for highly complex problems, whereas authority-based mechanisms become increasingly advantageous as complexity recedes (Nickerson & Zenger, 2004).

The KBV lens offers additional insight into why we observe a shift from hierarchical, vertically integrated firms toward ecosystem orchestration. Because actors are boundedly rational and cannot individually assimilate or process all relevant knowledge (Conner & Prahalad, 1996; Grant, 1996), they must choose between specializing deeply in a narrow domain or maintaining a broader, but more shallow orientation. But, contrary to the idiom that a jack-of-all-trades is a master of none, generalists in fact, often specialize in providing the technological or institutional "connective tissue" that enables communication and the exchange of otherwise disparate technical expert knowledge, thereby integrating ecosystem's various non-generically complementary parts and generating systemic outcomes. Without such integrators (e.g., lead firms) who maintain an overarching view of the problem and facilitate knowledge transfer, specialists may never connect or recognize their complementarities, as their expertise is often so narrowly focused that they overlook potential overlaps.[3] This is reflected in the ecosystem literature, where lead actors are frequently described as "facilitators" (Fu, Xiao, & Zhu, 2024), "platform leaders" (Helfat & Raubitschek, 2018) or "integrators" (Tabas, Nätti, & Komulainen, 2023), and where knowledge exchange often spans industry boundaries.

Finally, since the firm's knowledge can be seen as a "*portfolio of options, or platforms, on future developments*" (Kogut & Zander, 1992: 385), consolidation within ecosystems represents an even superior arrangement. By pooling their knowledge, ecosystem participants expand the collective option portfolio and, with it, the solution space available for tackling complex, high-value problems (Chintakananda, McIntyre, & Tong, 2025). Thus, ecosystems can *also* be viewed

[3] A similar point was raised by Henderson and Clark (1990) in their distinction between component knowledge and architectural knowledge.

as institutions for knowledge integration but may, in this respect, often surpass standalone firms because they combine the broad, generalist capabilities of platform leaders with the deep, specialist expertise of complementors, enabling wider and more diverse search. Just as firms historically served as foundations for coordinating knowledge flows between individuals, ecosystems now serve as foundations for knowledge orchestration between firms. In this sense, the pattern is recursive, yet the underlying objective of knowledge integration remains unchanged, making the KBV lens highly applicable to ecosystem research. By drawing out these parallels and explicit connections to codified and cognitive mechanisms, we argue that, going forward, the KBV and ecosystem literatures can more productively inform one another.

At this point, we should underscore that, given the multitude of concurrent mechanisms at play in ecosystem orchestration, no single theory can account for them alone. Ecosystem research will therefore inevitably continue, and indeed should continue, to rely on theoretical pluralism. In fact, many of these classic theories are themselves closely related and often complementary (Argyres, Felin, Foss, & Zenger, 2012; Gibbons, 2005; Zhang & Tong, 2021), and in combination they are better suited to explaining a phenomenon that operates at a wider unit of analysis than the firm or the bilateral inter-firm relationship. However, going forward, it is essential that future studies ground ecosystem research more firmly in established and highly applicable theoretical lenses. This requires future studies to be explicit about how particular theories map to the specific mechanisms or facets of orchestration they examine. We hope that, by distilling orchestration mechanisms and highlighting salient links to theories of the firm, our review will lead to more focused and theoretically grounded research, moving ecosystem and orchestration studies away from superficial, catch-all references while in turn sharpening and extending these foundational theories.

Finally, the theories we focus on here are not exhaustive. While we primarily outline theories that pertain to macro-level issues of organizational design and structure, micro-level theories related to leadership, trust, and managerial cognition are equally important and could further advance our understanding of ecosystem orchestration. In the discussion that follows, we expound on this and several other notions, suggesting multiple paths through which ecosystem research can be advanced.

ADVANCING ECOSYSTEM RESEARCH

We began our article with the notion that firms' control extends beyond their formal boundaries, particularly in an increasingly digitally connected economy. At one time, this led scholars to pose the question of how firms can orchestrate the activities of other organizations without fiat (Dhanaraj & Parkhe, 2006). Yet, despite nearly two decades of research directly or indirectly addressing this question, the field has also produced divergent streams and a fragmented body of literature, with unclear connections between ecosystem types and little clarity on how they relate to or can advance established theories of the firm. As previous studies noted, this has led some scholars to question the concept's value and dismiss ecosystems as a catch-all term (c.f. Autio, 2022; Burström et al., 2024; Thomas & Autio, 2020).

Having said that, we undertook a review of ecosystem orchestration to integrate findings on both different ecosystem types and the mechanisms that lead firms deploy to exert such influence, and to advance a typology of orchestration regimes—a heuristic tool to help clarify how these mechanisms are developed, combined, and influenced. In doing so, we were also able to draw out several connections to established theories of the firm—connections that are commonly overlooked—thereby seeking to anchor the ecosystem concept more firmly within them. We now turn to elaborate these connections and outline directions for future research and key implications.

Grounding ecosystem research in established theories. A major shortcoming of the ecosystem unit of analysis has been its unclear boundaries and its vague connections to established theories of the firm—a concern that has already been noted in prior reviews on platforms, which called for research to bridge this divide (Chen et al., 2022; Rietveld & Schilling, 2021). By extracting the mechanisms that consistently appear across the ecosystem orchestration literature, we identified explicit links to several classic theories of the firm and to distinct ecosystem types, thus providing a clearer foundation for integrating ecosystem research with mainstream theoretical traditions. Researchers should recognize that many of the mechanisms we have identified in our review have, in fact, been discussed within their respective literature streams long before the ecosystem concept emerged. While most ecosystem studies do not draw on classic theories—resulting in limited theoretical grounding—those that do typically approach orchestration through the RBV and related resource-orchestration literature. This lens is widely applicable because it does not explicitly concern itself with any specific mechanism, but instead relates to them all at a more abstract level. However, although the RBV has been applied to ecosystem research, it has not itself been revisited by scholars in light of the ecosystem phenomenon. We therefore encourage future work to revisit RBV and the VRIN framework (Barney, 1991, 2001), particularly as competitive advantage increasingly stems from accessing externally located resources and orchestrating and integrating them across organizational boundaries via the mechanisms we identify. Likewise, although prior work has begun to articulate how TCE relates to ecosystem orchestration (Foss et al., 2023; Jacobides et al., 2018), further theorizing is needed to situate ecosystems more clearly along the continuum between market-based externalization and hierarchical internalization.

More importantly, while TCE and RBV appear with some regularity in ecosystem research, KBV and PRT remain notably underrepresented, receiving only implicit or cursory mention. Although cognitive and codified mechanisms are central to both KBV and ecosystem research and appear consistently in the studies we reviewed, this connection has not previously been articulated. To date, KBV scholars have not applied the lens to explain ecosystem phenomena, nor have ecosystem scholars revisited KBV by integrating insights from ecosystem research. Future studies can build on the connections we identify between ecosystem types and KBV-related mechanisms to examine how knowledge codification and reliance on cognitive mechanisms evolve in various ecosystems. For example, by conceptualizing how codification enables the modularization of interdependent knowledge exchanges, KBV could help clarify why such interdependencies become more pooled in ecosystems rather than sequential, as in supply chains or other bilateral arrangements. Likewise, PRT has been seldom considered in ecosystem studies, despite its clear relevance to economic-incentive mechanisms and notions of ownership (Chen et al., 2022). By jointly drawing on both PRT and RBV, there is potential to advance theory of ecosystems by conceptualizing orchestration as the regulation of access to critical external resources. While our mechanism-to-theory mapping does not claim to establish exclusive links, it brings the most salient connections to the fore. In doing so, it outlines promising pathways for grounding ecosystem studies more firmly in classic theories of the firm and, in turn, for advancing those theories through insights generated from an ecosystem perspective.

Improving the connection of ecosystem types to orchestration mechanisms. We find that researchers often do not take sufficient care in distinguishing between ecosystem sub-types, applying labels such as “innovation ecosystem” or “entrepreneurial ecosystem” rather casually, without specifying what sets them apart. Our integrative review indicates that the ecosystem literature can be organized around four core sub-types—innovation, entrepreneurial, platform, and industry ecosystems—each distinguished by its orientation and the orchestration regime mobilized to support it. Ecosystems oriented toward innovation and entrepreneurial search, and thus toward exploration, typically depend more on cognitive and incentive-based mechanisms. In contrast, platform and industrial ecosystems, which prioritize more immediate commercialization and thus exploitation, tend to rely predominantly on codified and hierarchical mechanisms. In these settings, a well-established industry structure or dominant platform architecture reduces uncertainty around design and exchange rules, enabling higher degrees of codification. Moreover, innovation and entrepreneurial ecosystems are quite often regional and tied to a specific location, as ongoing close interaction, reliance on cognitive mechanisms, and the exchange of tacit, uncodifiable knowledge frequently necessitate geographic proximity. Platform and industrial ecosystems, although often tailored to a particular market, are less geographically constrained. Furthermore, while additional types such as circular, knowledge, or service ecosystems exist, these generally fall under aforementioned broader categories. We therefore encourage future research to position ecosystems within these overarching sub-types rather than introduce new labels. To distinguish across the existing sub-types, we propose focusing on the underlying orientation (exploration vs. exploitation) and the most salient orchestration regime as illustrated by our typology in Figure 1.

Engaging with public administration research. Not only do ecosystems cross industry boundaries (Shi et al., 2023), they also span private and public domains. Our review finds that ideas resembling ecosystem orchestration appear in public management and administration, where scholars likewise emphasize mutual reliance and interdependence among stakeholder groups and note that the role of governments and public actors has become less directive and more facilitative (Ansell & Torfing, 2021; Lebec & Dudau, 2024). However, while sharing broadly similar units of analysis to that of ecosystems and orchestration, scholars in the public administration domain—coming from a different intellectual tradition—typically employ different terminology, such as "collaborative governance" (Kinder et al., 2021) and "co-creation" (Ansell & Torfing, 2021), which partly explains why the two fields remain somewhat siloed and rarely exchange insights. Yet as the two disciplines increasingly view both public and private entities as potential ecosystem actors or orchestrators, the line between public administration and management has grown progressively more porous, creating meaningful opportunities for the two literatures to inform one another in relation to ecosystem studies. For instance, *administrative control*—which appears underrepresented in the management literature relative to other orchestration regimes—could potentially be informed by insights from public management scholarship, helping to address this gap. Likewise, our review offers a foundation for public management scholars seeking to incorporate ecosystem insights from the management field. Moving forward, research in both domains would benefit from greater awareness of these intersections and more deliberate cross-disciplinary engagement to support ongoing cross-fertilization.

Sharpening the boundaries of the ecosystem and orchestration constructs. In connection to the above, researchers should carefully assess, in each case, whether public actors and government bodies ought to be treated as ecosystem participants or merely as external regulators. The literature often exhibits confusion on this point, with ecosystem boundaries left unspecified and little justification for why passive government entities are sometimes classified as ecosystem members. Simon notes that (1996: 87) "*there is a certain arbitrariness in drawing the boundary between inner and outer environments of artificial systems.*" This is certainly true for ecosystems, which are inherently open systems in which actors and their activities are interdependent to varying and shifting degrees (Shipilov & Gawer, 2020)—not only internally but also in relation to the broader external environment—making the delineation of boundaries an analytical rather than an ontological issue. Because this arbitrariness can in turn make the ecosystem concept a convenient catch-all term for describing nearly any organizational configuration and thus dilute its value, the field needs clearer, shared convention among ecosystem researchers. Some earlier works have contributed to such conventions, building a clearer sense that ecosystem boundaries tend to encompass groups of multilateral actors brought together by a common value proposition (Adner, 2017), who share some form of non-generic complementarity which requires coordination across firms (Jacobides et al., 2018; Shipilov & Gawer, 2020), and where control is internal to orchestrators while value creation and capture activities are predominantly external to them (Altman et al., 2022). Accordingly, public—or any other—actors should be assessed along these dimensions to determine whether they are ecosystem participants or part of the external regulatory environment. We suggest that when public actors play a passive role, acting solely in a regulative capacity within a given institutional environment, they should be treated as part of ecosystem's external context. However, when a public actor becomes an active participant—orchestrating ecosystem activities or actively complementing to a value proposition of a focal ecosystem, rather than the institutional field at large—it may legitimately be regarded as an ecosystem actor. If this is the case, authors should clearly articulate and convincingly demonstrate why they classify a particular actor, public or otherwise, as an ecosystem actor rather than part of its external context, grounding their reasoning in a careful assessment of the ecosystem's locus of control and locus of activity (Altman et al., 2022), as well as specifying the orchestration mechanisms and regime at play. We believe this clearer distinction will also help prevent conflating orchestration with governance, regulation and coordination—a distinction that is often ambiguous in the literature. In our view, orchestration denotes a specific form of coordination in which a central actor strategically bundles and deploys codified, cognitive, incentive-based, and hierarchical control mechanisms to direct and influence activities that lie beyond its organizational boundaries. Future research should remain attentive to these distinctions so as not to further dilute the already intricate ecosystem and orchestration constructs.

Differentiating and generalizing across external contexts. Having said that, when government actors are not part of a focal ecosystem, they still constitute an external, nonmarket context that can significantly shape how the ecosystem is orchestrated. For instance, an ecosystem leader and its standards may ultimately gain formal recognition by government bodies (e.g., Fu et al., 2024; Mozheiko, 2025; Zhang, Li, & Tong, 2024), thereby reinforcing the lead firm's central position. Institutional environments, market orientations, and local cultural conditions likewise matter. For example, trust-based mechanisms may become especially important in emerging markets with weak institutional infrastructures (Zeng et al., 2021). Furthermore, despite global digital connectivity, geopolitical polarization increasingly influences internationalization strategies and decisions to multihome—particularly in AI ecosystems, which carry strategic significance for nation states (Zhou et al., 2025). Overall, although ecosystem research has gained some traction within international business studies, it still remains limited in explaining how orchestration practices vary across external contexts and how these contextual differences shape orchestration regimes. There is significant opportunity for future research to conduct comparative analyses of orchestration regimes across different regions, institutional settings, and geopolitical contexts. In this connection, future work should examine lead firms' nonmarket strategies by shifting attention beyond the internal orchestration of ecosystem participants to external orchestration—that is, how and through what mechanisms ecosystem actors strategically exert influence on politicians, regulators, and the general public.

Expanding methodological diversity. In our review sample, the majority of papers were qualitative case studies, along with some conceptual works and earlier reviews on adjacent ecosystem topics. Surprisingly, quantitative research on ecosystem orchestration remains very limited—likely due to the difficulty of operationalization and the lack of clarity about what exactly could be measured. By outlining codification and centralization as the two central dimensions shaping the choice of mechanisms and regimes, as well as showing how ecosystem types vary to some extent depending on whether they are geared toward explorative or exploitative objectives, we provide a point of departure for potential future quantitative and mixed-methods research designs on ecosystem orchestration. We also find that little ethnographic, shadowing, and participant-observation fieldwork is present in the literature, which reflects our observation that very few studies in ecosystem contexts address micro- (individual) and meso- (group) levels of analysis. Strategic choices rest on the judgments of individual managers and teams. Consequently, individual-level differences should inevitably lead to variation in orchestration regimes and make-buy decisions. Cognitive mechanisms, for instance, are not readily available but must be cultivated, making leadership a vital component of ecosystem orchestration. For a shared vision to form, it must be communicated by a capable and influential leader (Foss et al., 2023)—a capability that is likely to vary across ecosystems. Furthermore, firms function as platforms for shared culture and cognitive frames, where individuals can develop a strong sense of affiliation and unity with their organization (Ashforth & Mael, 1989). Yet how does this sense of an individual's affiliation evolve within ecosystems, where organizational boundaries and tasks become increasingly blurred, and relational ties extend across firms? And how do leaders cultivate shared mental models and a common vision among individuals when realizing a focal value proposition depends on ecosystem-level orchestration rather than a single firm? Although research frequently references cognitive and relational mechanisms, these mentions remain cursory, leaving many questions unanswered, and the managerial cognition and micro-level theories have yet to engage more fully with ecosystem research. Moving beyond a largely case-based research tradition to encompass a broader range of methodologies and levels of analysis would deepen our understanding of variance in ecosystem orchestration performance, as well as the competitive and collaborative dynamics that shape it on different levels.

Deepening understanding of ecosystem-level competitive dynamics. A firm's competitive advantage derives from a configuration of activities that are complementary and mutually reinforcing (Porter, 1996). In today's economy, however, these complementary activities are no longer primarily internal to the firm but are, to a considerable extent external. Developing and orchestrating an ecosystem can enable firms to outperform their standalone counterparts by giving them access to a broader pool of complementary, externally situated resources, capabilities, and knowledge. Competitive dynamics have thus become more intricate, as firms can create value in collaboration with competitors, operate under dual business models, and multihome as complementors simultaneously across competing ecosystems. This expanding web of interdependence fundamentally reshapes organizational and strategic thinking. The locus of competition has progressively shifted—from firms, to alliances, to platforms, and now to ecosystems—with each step broadening the strategic arena while intensifying interdependence. Although this progression follows a recursive pattern, the scope of strategic choice has widened considerably. The implication for firms is that they must now decide not only how to compete, but also with whom to collaborate, how to orchestrate diverse external complementors, and how to position themselves as ecosystem actors rather than standalone organizations, while preserving a measure of independence. Questions of ecosystem positioning and multihoming as an internationalization strategy have become central concerns, amid heightened geopolitical uncertainty (Zhou et al., 2025). Consequently, competitive advantage is no longer sustained through isolated action but through collaborative advantage, rooted in effective ecosystem positioning and in the capability to deploy orchestration regimes that mobilize and integrate externally distributed resources.

Yet, although research on platform competition has advanced and been reviewed (Rietveld & Schilling, 2021), studies on ecosystem competitive dynamics still remain heavily skewed toward "born-digital" ecosystems. Far less attention has been paid to, for example, the antecedents that prompt successful incumbent standalone firms to pivot toward an ecosystem strategy, how ecosystem leaders subsequently outperform standalone firms within the same industry, or, conversely, the circumstances in which standalone incumbent firms outcompete ecosystems. Systematic empirical comparisons between ecosystem leaders and integrated hierarchies would meaningfully advance theoretical discussions on how complementarities and asset specificity shape make-or-buy choices (Argyres & Zenger, 2012; Baldwin, 2008; Jacobides et al., 2018), while deepening our understanding of historically novel organizational forms—ecosystems in which effective control is exercised without ownership over external complementors, yet still achieves the integration of their distributed effort.

CONCLUSION

In an increasingly connected and interdependent economy, firms' competitive advantage often stems from complementarities that lie outside their boundaries. As a result, they must consider how to influence and manage activities, knowledge, and resource flows beyond what they own. To do so, they combine cognitive, codified, incentive-based, and hierarchical mechanisms into orchestration regimes suited to integrating resources and knowledge dispersed across firms—

whether the aim is exploration in innovation and entrepreneurial ecosystems or exploitation in platform and industrial ecosystems. While ecosystems themselves are relatively new organizational forms, the underlying orchestration mechanisms are not, and are in fact deeply rooted in classic theories of the firm. It is through these mechanisms that ecosystem studies can be brought into closer alignment with established theories of the firm, and in turn, help refine these theories through insights gleaned from the ecosystem lens.

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Table 1
Related Reviews

Prior reviews	Ecosystem focus	Focus on mechanisms
Altman et al. (2022)	All types	Peripheral
Chen et al. (2022)	Digital platforms	Strong
Kolagar et al. (2022)	Digital ecosystems	Peripheral
Rietveld & Schilling (2021)	Platform ecosystems	Peripheral
Shi et al. (2023)	All types	Peripheral
Shipilov & Gawer (2020)	Ecosystems and networks	Peripheral

Table 2
Orchestration Mechanisms

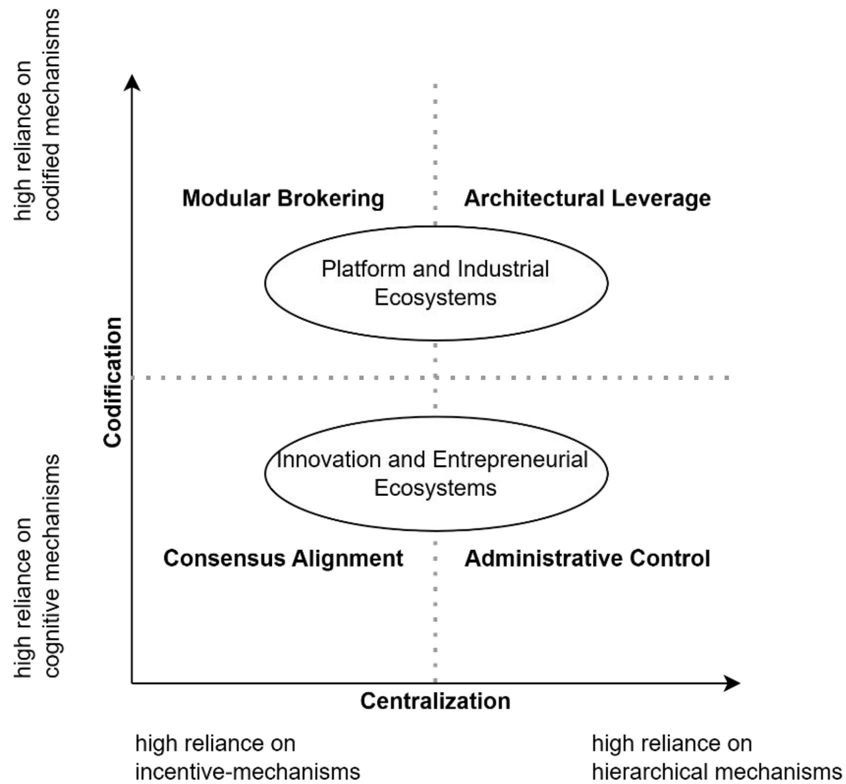
Orchestration Mechanism	Definition	Examples	Representative Publications
Codified mechanisms	A tangible, device or interface that encapsulates information and functions as a shared knowledge repository and reference point, reducing ongoing one-to-one exchanges and buffering participants from coordination burdens.	APIs, SDKs, technical standards, contracts, databases, algorithms, AI tools, etc.	Jacobides et al., 2016; Jensen et al., 2024; Reiter et al., 2024; Sjödin et al., 2022; Stonig et al., 2022
Cognitive mechanisms	An intangible device that, despite its tacit and uncodifiable nature, functions to ensure alignment and coherence in the behavior of ecosystem members.	Mental models, cognitive frames, trust, narratives, collective identities, informal norms, social ties, social relationships, culture, etc.	Cennamo et al., 2022; Fang et al., 2021; Hernández-Chea et al., 2021; Reypens et al., 2021; Shi et al., 2024
Incentive-mechanisms	Instruments that convincingly signal greater value from participating in an ecosystem than an actor could capture independently.	Network effects, subsidies, value perception, pricing strategies, nonpecuniary incentives, access to greater extent of the market, etc.	Lu & Zhang, 2024; Masucci et al., 2020; Mozheiko, 2025; Savaget et al., 2024; Tavalaei et al., 2025
Hierarchical mechanisms	Instruments that enable quasi-hierarchical control, usually through power asymmetries and control of nonfungible key assets.	Bargaining power, switching costs, central network position, formal alliances, governance	Kude & Huber, 2025; Laczko et al., 2019; Mérindol & Versailles, 2024;

		structures, IP rights, etc.	Zhang et al., 2022; Zhu & Liu, 2018

Table 3
Orchestration Regimes

Regime	Primary Mechanisms	Definition	Typical Ecosystem types
Architectural Leverage	Codified and hierarchical	An orchestration regime defined by codified coordination and high centralization of power.	Platform ecosystems, Industrial ecosystems
Modular Brokering	Codified and incentive	An orchestration regime defined by codified coordination and low concentration of power.	Platform ecosystems, Industrial ecosystems
Administrative Control	Cognitive and hierarchical	An orchestration regime defined by cognitive coordination and high concentration of power.	Innovation ecosystems, entrepreneurial ecosystems
Consensus Alignment	Cognitive and incentive	An orchestration regime defined by cognitive coordination and low concentration of power.	Innovation ecosystems, entrepreneurial ecosystems

Figure 1
Typological Framework of Orchestration Regimes



Appendix A
Audit Trail